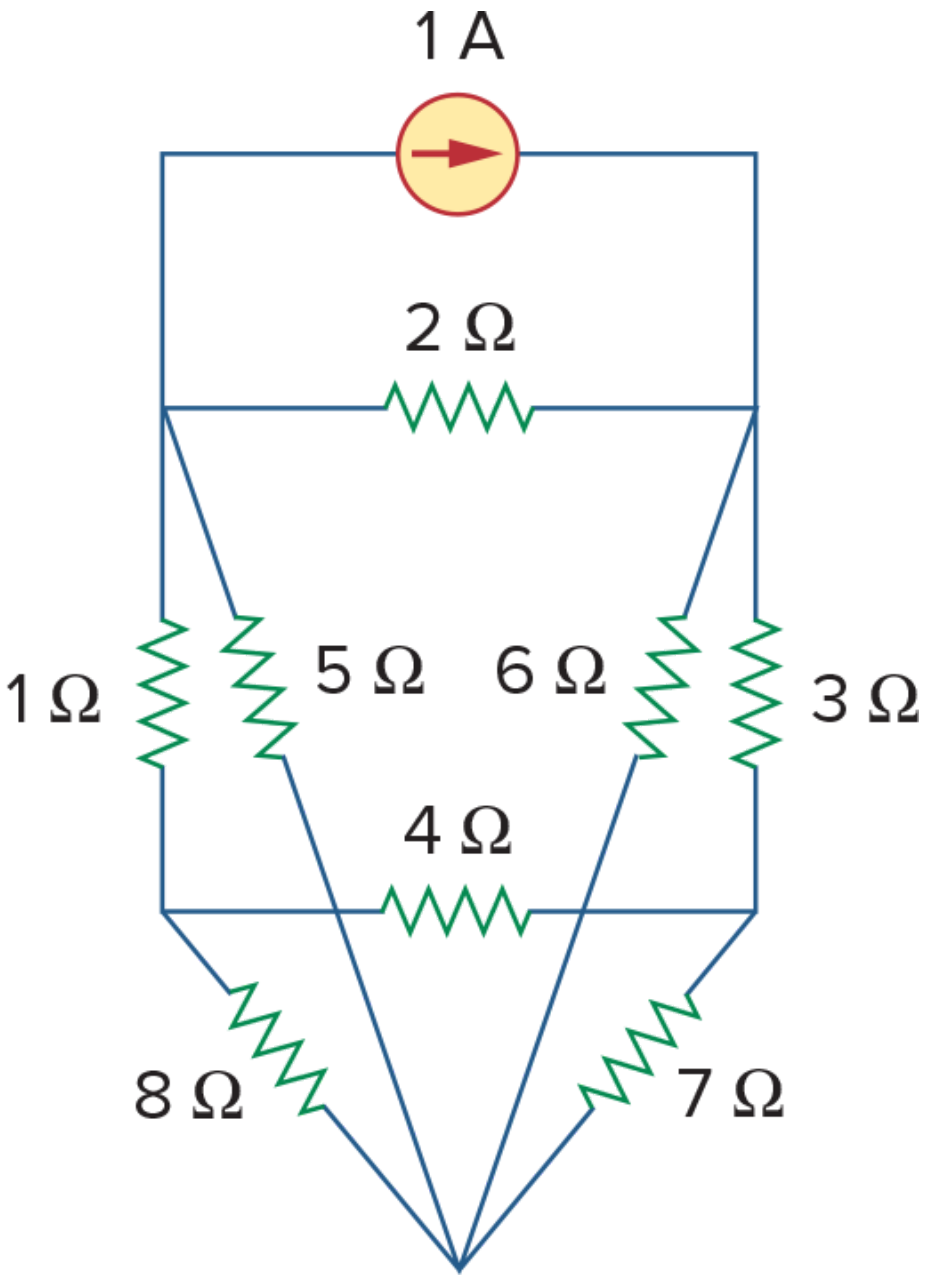
# 3.4 Mesh Analysis

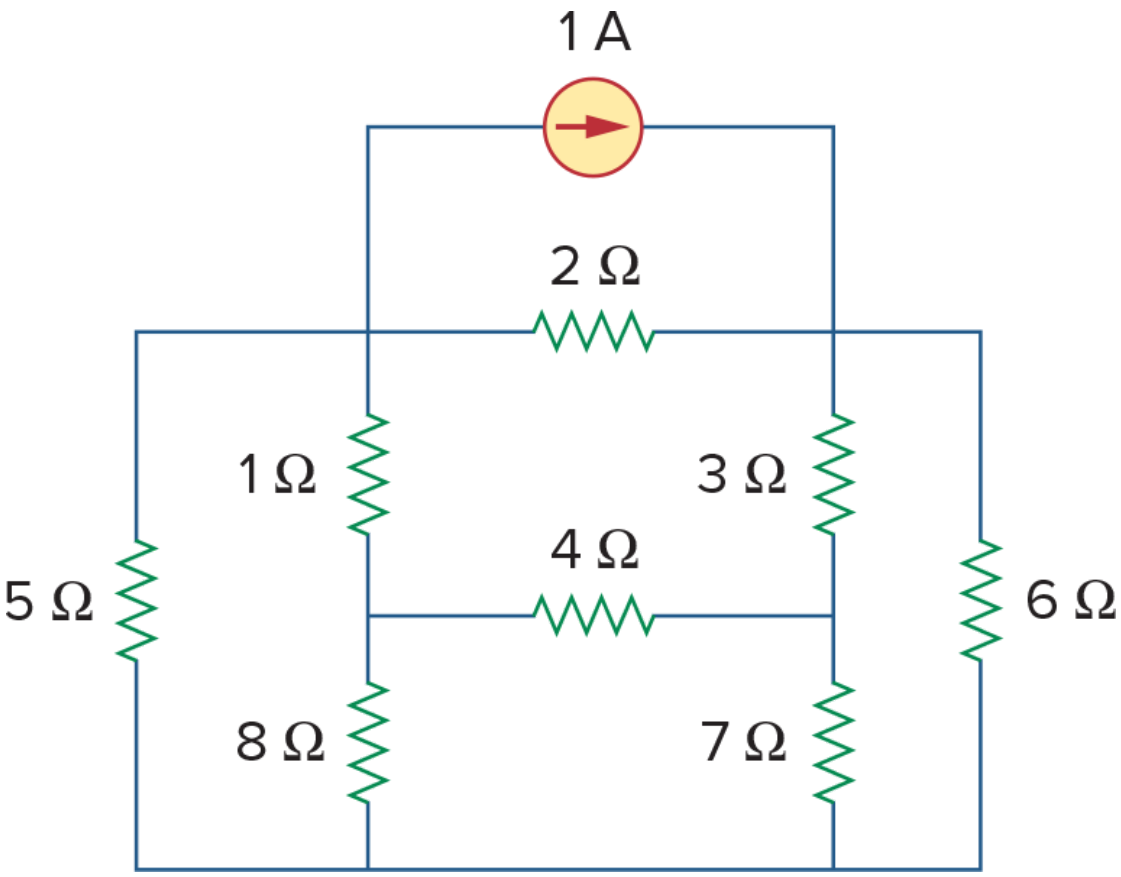
Mesh analysis provides another general procedure for analyzing circuits, using mesh currents as the circuit variables. Using mesh currents instead of element currents as circuit variables is convenient and reduces the number of equations that must be solved simultaneously. Recall that a loop is a closed path with no node passed more than once. A mesh is a loop that does not contain any other loop within it.

Mesh analysis is also known as *loop analysis* or the *mesh-current method*.

Nodal analysis applies KCL to find unknown voltages in a given circuit, while mesh analysis applies KVL to find unknown currents. Mesh analysis is not quite as general as nodal analysis because it is only applicable to a circuit that is *planar*. A planar circuit is one that can be drawn in a plane with no branches crossing one another; otherwise it is *nonplanar*. A circuit may have crossing branches and still be planar if it can be redrawn such that it has no crossing branches. For example, the circuit in Fig. 3.15(a) has two crossing branches, but it can be redrawn as in Fig. 3.15(b). Hence, the circuit in Fig. 3.15(a) is planar. However, the circuit in Fig. 3.16 is nonplanar, because there is no way to redraw it and avoid the branches crossing. Nonplanar circuits can be handled using nodal analysis, but they will not be considered in this text.




(a)




(b)

Figure 3.15 (a) A planar circuit with crossing branches, (b) the same circuit redrawn with no crossing branches.

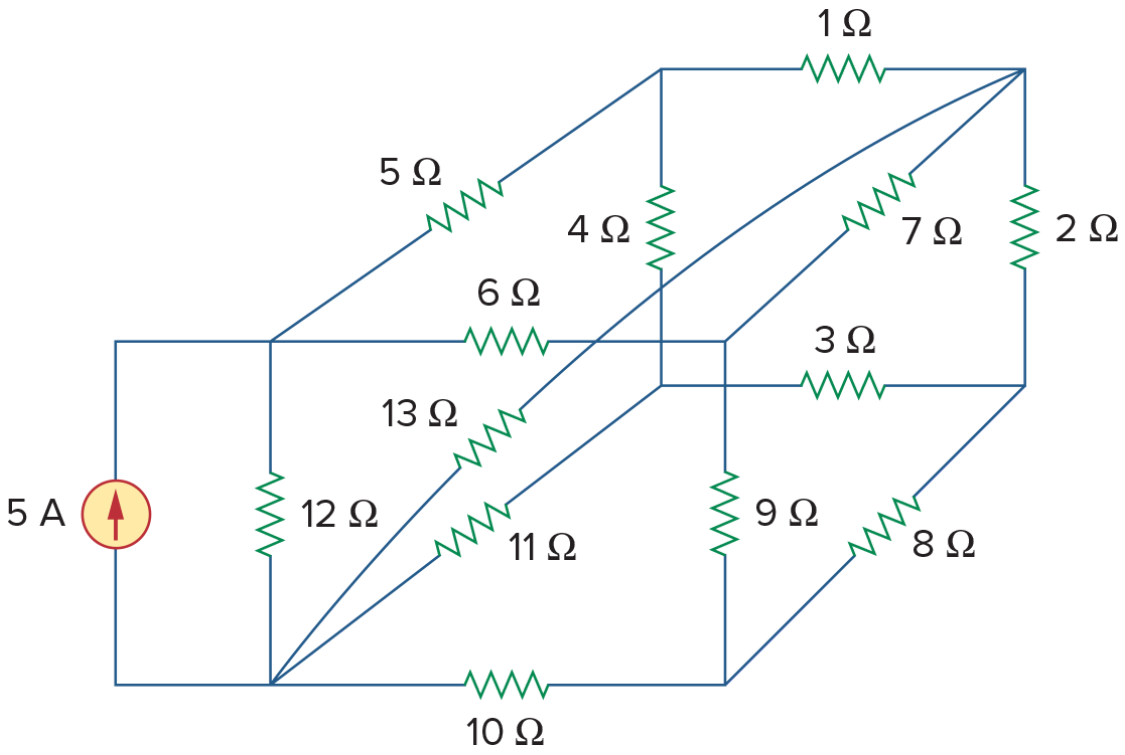



Figure 3.16 A nonplanar circuit.

To understand mesh analysis, we should first explain more about what we mean by a mesh.

A **mesh** is a loop that does not contain any other loops within it.

Page 92In Fig. 3.17, for example, paths *abefa* and *bcdeb* are meshes, but path *abcdefa* is not a mesh. The current through a mesh is known as *mesh current*. In mesh analysis, we are interested in applying KVL to find the mesh currents in a given circuit.

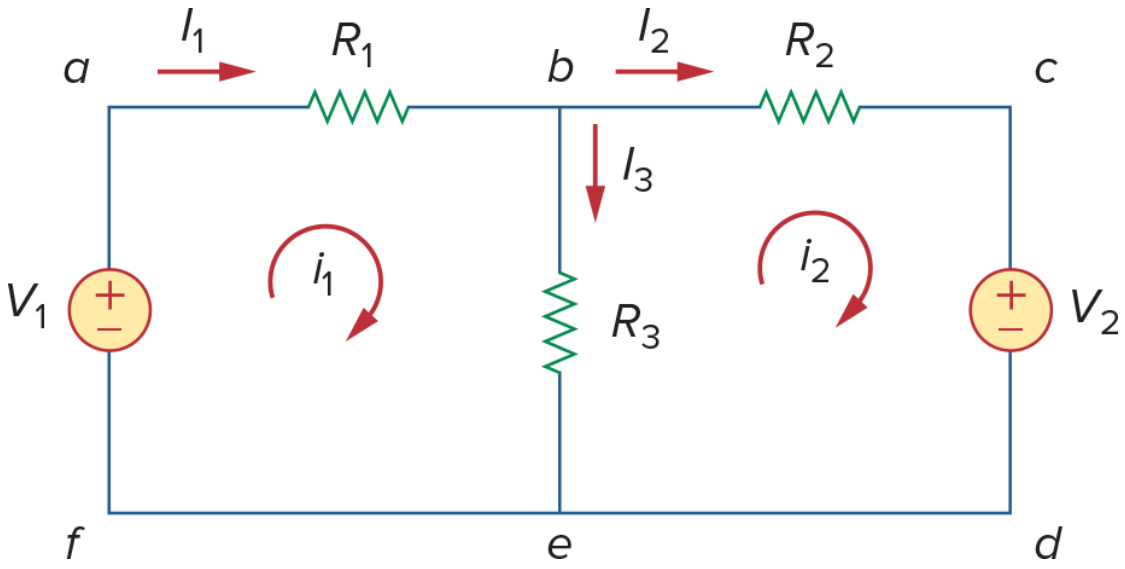



Figure 3.17 A circuit with two meshes.

Although path *abcdefa* is a loop and not a mesh, KVL still holds. This is the reason for loosely using the terms *loop analysis* and *mesh analysis* to mean the same thing.

In this section, we will apply mesh analysis to planar circuits that do not contain current sources. In the next section, we will consider circuits with current sources. In the mesh analysis of a circuit with *n* meshes, we take the following three steps.

**Steps to Determine Mesh Currents:**

1. Assign mesh currents $i_1, i_2, \ldots, i_n$ to the *n* meshes.
2. Apply KVL to each of the *n* meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting *n* simultaneous equations to get the mesh currents.

The direction of the mesh current is arbitrary—(clockwise or counterclockwise)—and does not affect the validity of the solution.

To illustrate the steps, consider the circuit in Fig. 3.17. The first step requires that mesh currents $i_1$ and $i_2$ are assigned to meshes 1 and 2. Although a mesh current may be assigned to each mesh in an arbitrary direction, it is conventional to assume that each mesh current flows clockwise.

As the second step, we apply KVL to each mesh. Applying KVL to mesh 1, we obtain

$$-V_1 + R_1 i_1 + R_3(i_1 - i_2) = 0$$

or

$$(R_1 + R_3)i_1 - R_3 i_2 = V_1 \quad \textbf{(3.13)}$$

For mesh 2, applying KVL gives

$$R_2 i_2 + V_2 + R_3(i_2 - i_1) = 0$$

or

$$-R_3 i_1 + (R_2 + R_3)i_2 = -V_2 \quad \textbf{(3.14)}$$

Note in Eq. (3.13) that the coefficient of $i_1$ is the sum of the resistances in the first mesh, while the coefficient of $i_2$ is the negative of the resistance common to meshes 1 and 2. Now observe that the same is true in Eq. (3.14). This can serve as a shortcut way of writing the mesh equations. We will exploit this idea in Section 3.6.

The shortcut way will not apply if one mesh current is assumed clockwise and the other assumed counter-clockwise, although this is permissible.

Page 93The third step is to solve for the mesh currents. Putting Eqs. (3.13) and (3.14) in matrix form yields

$$\begin{bmatrix} R_1 + R_3 & -R_3 \\ -R_3 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix} \quad \textbf{(3.15)}$$

which can be solved to obtain the mesh currents $i_1$ and $i_2$. We are at liberty to use any technique for solving the simultaneous equations. According to Eq. (2.12), if a circuit has *n* nodes, *b* branches, and *l* independent loops or meshes, then $l = b - n + 1$. Hence, *l* independent simultaneous equations are required to solve the circuit using mesh analysis.

Notice that the branch currents are different from the mesh currents unless the mesh is isolated. To distinguish between the two types of currents, we use *i* for a mesh current and *I* for a branch current. The current elements $I_1$, $I_2$, and $I_3$ are algebraic sums of the mesh currents. It is evident from Fig. 3.17 that

$$I_1 = i_1, \qquad I_2 = i_2, \qquad I_3 = i_1 - i_2 \quad \textbf{(3.16)}$$

**Example 3.5**

For the circuit in Fig. 3.18, find the branch currents $I_1$, $I_2$, and $I_3$ using mesh analysis.

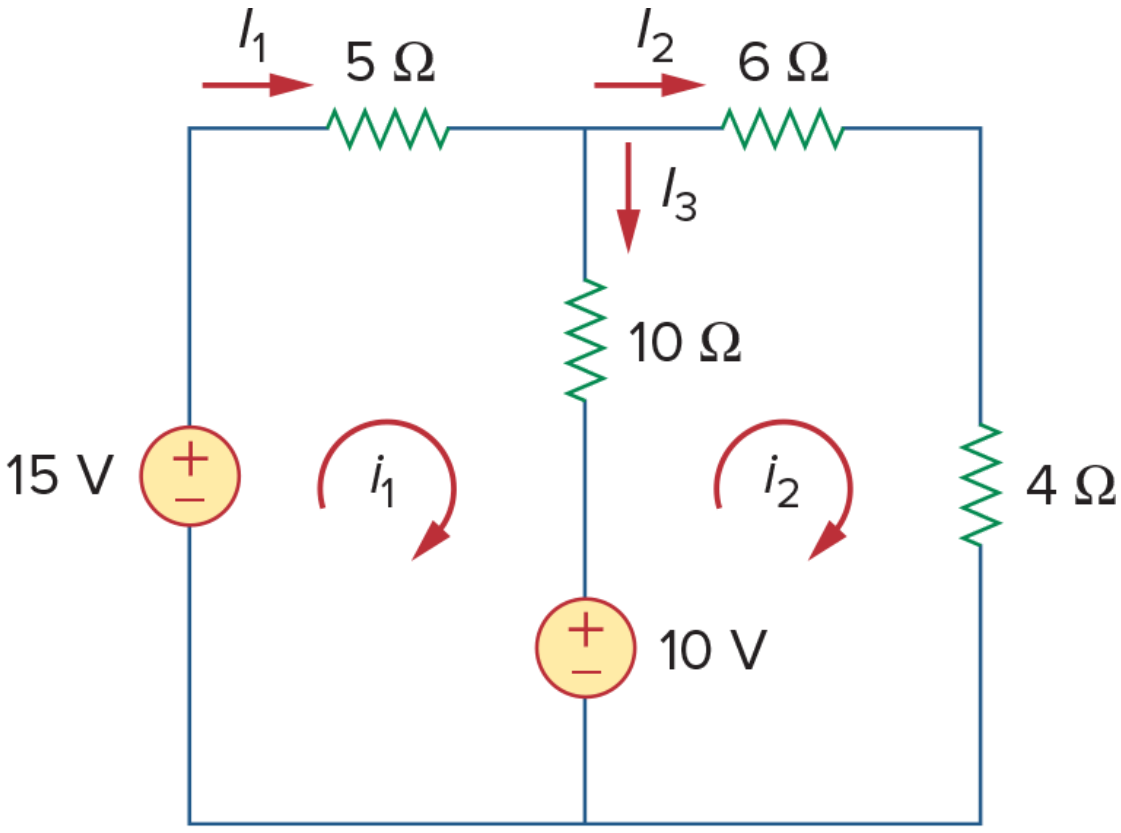



Figure 3.18 For Example 3.5.

**Solution:**

We first obtain the mesh currents using KVL. For mesh 1,

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

or

$$3i_1 - 2i_2 = 1 \quad \textbf{(3.5.1)}$$

For mesh 2,

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

or

$$i_1 = 2i_2 - 1 \quad \textbf{(3.5.2)}$$

▪ METHOD 1 Using the substitution method, we substitute Eq. (3.5.2) into Eq. (3.5.1), and write

$$6i_2 - 3 - 2i_2 = 1 \qquad \Rightarrow \qquad i_2 = 1 \text{ A}$$

From Eq. (3.5.2), $i_1 = 2i_2 - 1 = 2 - 1 = 1$ A. Thus,

$$I_1 = i_1 = 1 \text{ A}, \qquad I_2 = i_2 = 1 \text{ A}, \qquad I_3 = i_1 - i_2 = 0$$

▪ METHOD 2 To use Cramer's rule, we cast Eqs. (3.5.1) and (3.5.2) in matrix form as

$$\begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Page 94We obtain the determinants

$$\Delta = \begin{vmatrix} 3 & -2 \\ -1 & 2 \end{vmatrix} = 6 - 2 = 4$$

$$\Delta_1 = \begin{vmatrix} 1 & -2 \\ 1 & 2 \end{vmatrix} = 2 + 2 = 4, \qquad \Delta_2 = \begin{vmatrix} 3 & 1 \\ -1 & 1 \end{vmatrix} = 3 + 1 = 4$$

Thus,

$$i_1 = \frac{\Delta_1}{\Delta} = 1 \text{ A}, \qquad i_2 = \frac{\Delta_2}{\Delta} = 1 \text{ A}$$

as before.

**Practice Problem 3.5**

Calculate the mesh currents $i_1$ and $i_2$ of the circuit of Fig. 3.19.

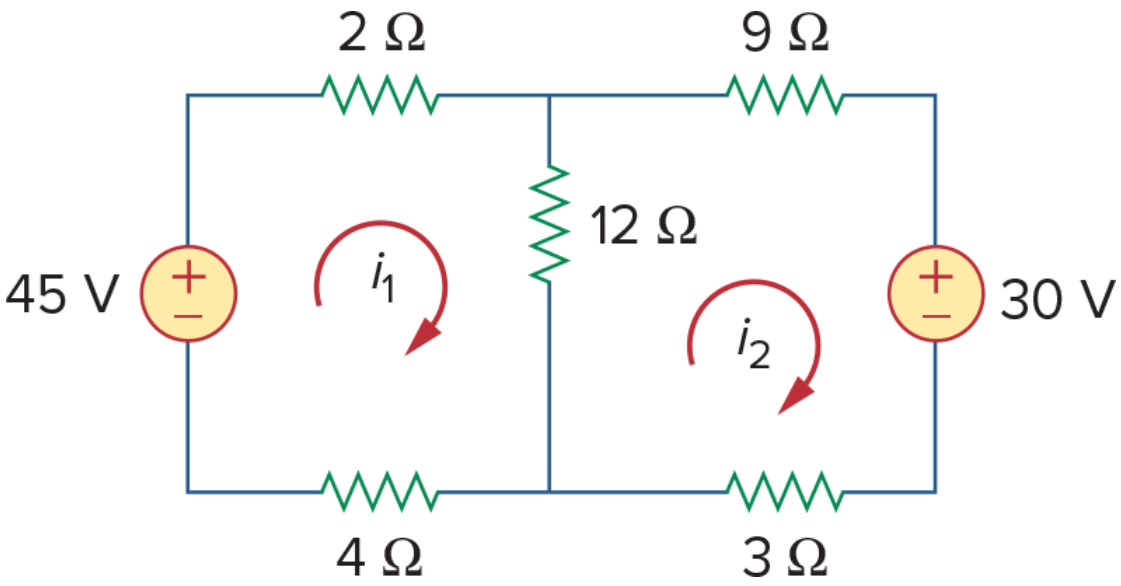



Figure 3.19 For Practice Prob. 3.5.

Answer: $i_1 = 2.5$ A, $i_2 = 0$ A.

**Example 3.6**

Use mesh analysis to find the current $I_o$ in the circuit of Fig. 3.20.

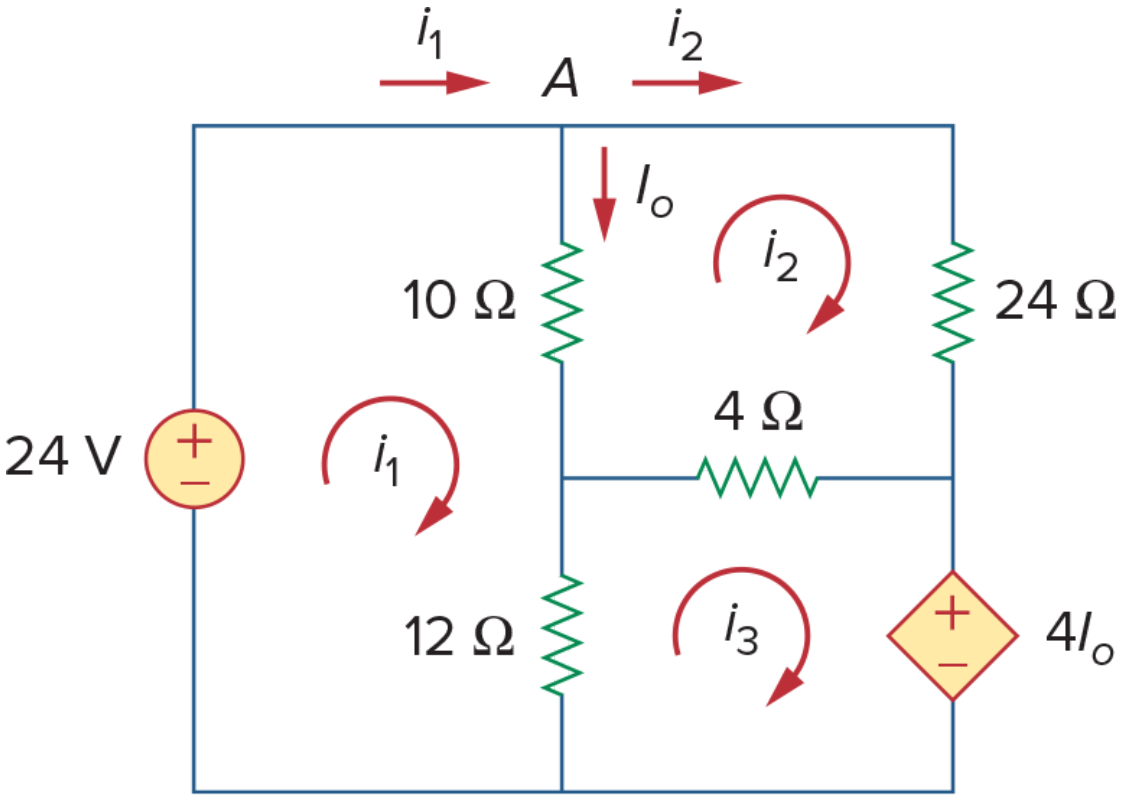



Figure 3.20 For Example 3.6.

**Solution:**

We apply KVL to the three meshes in turn. For mesh 1,

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$$

or

$$11i_1 - 5i_2 - 6i_3 = 12 \quad \textbf{(3.6.1)}$$

For mesh 2,

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0$$

or

$$-5i_1 + 19i_2 - 2i_3 = 0 \quad \textbf{(3.6.2)}$$

For mesh 3,

$$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

Page 95But at node A, $I_o = i_1 - i_2$, so that

$$4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

or

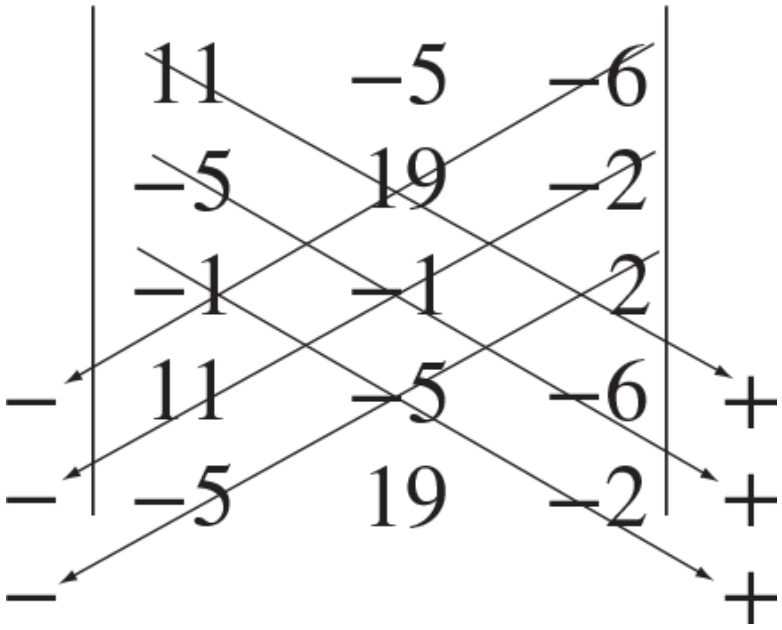
$$-i_1 - i_2 + 2i_3 = 0 \textbf{ (3.6.3)}$$

In matrix form, Eqs. (3.6.1) to (3.6.3) become

$$\begin{bmatrix} 11 & -5 & -6 \\ -5 & 19 & -2 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$

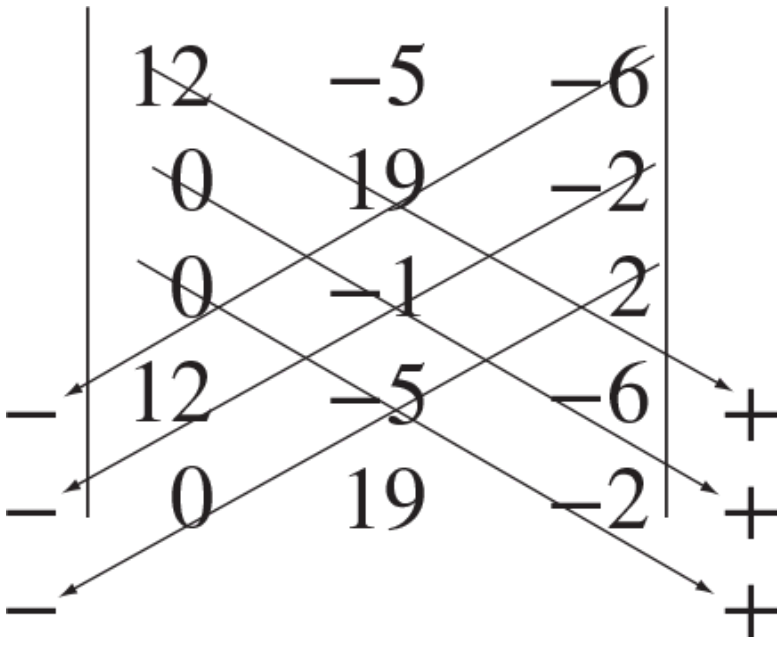
We obtain the determinants as

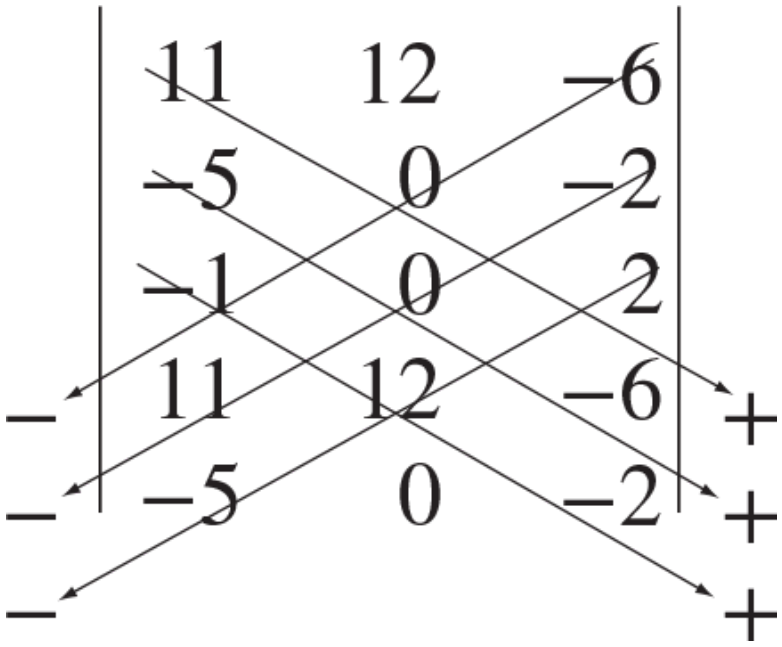
$\Delta =$



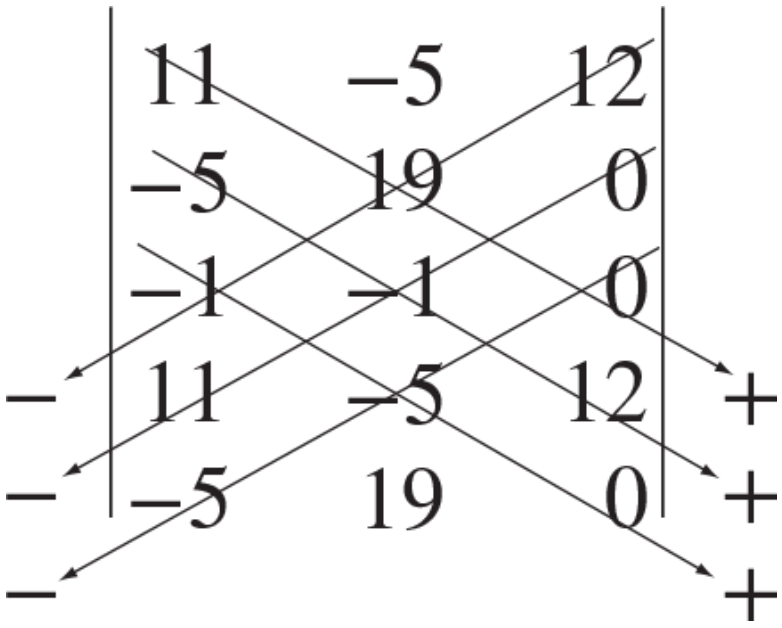

$= 418 - 30 - 10 - 114 - 22 - 50 = 192$

$\Delta_1 =$ $= 456 - 24 = 432$




$\Delta_2 =$  $= 24 + 120 = 144$


$\Delta_3 =$ $= 60 + 228 = 288$




We calculate the mesh currents using Cramer's rule as

$$i_1 = \frac{\Delta_1}{\Delta} = \frac{432}{192} = 2.25\ \text{A}, \qquad i_2 = \frac{\Delta_2}{\Delta} = \frac{144}{192} = 0.75\ \text{A},$$

$$i_3 = \frac{\Delta_3}{\Delta} = \frac{288}{192} = 1.5\ \text{A}$$

Thus, $I_o = i_1 - i_2 = 1.5\ \text{A}$.

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**Practice Problem 3.6**

Using mesh analysis, find $I_o$ in the circuit of Fig. 3.21.

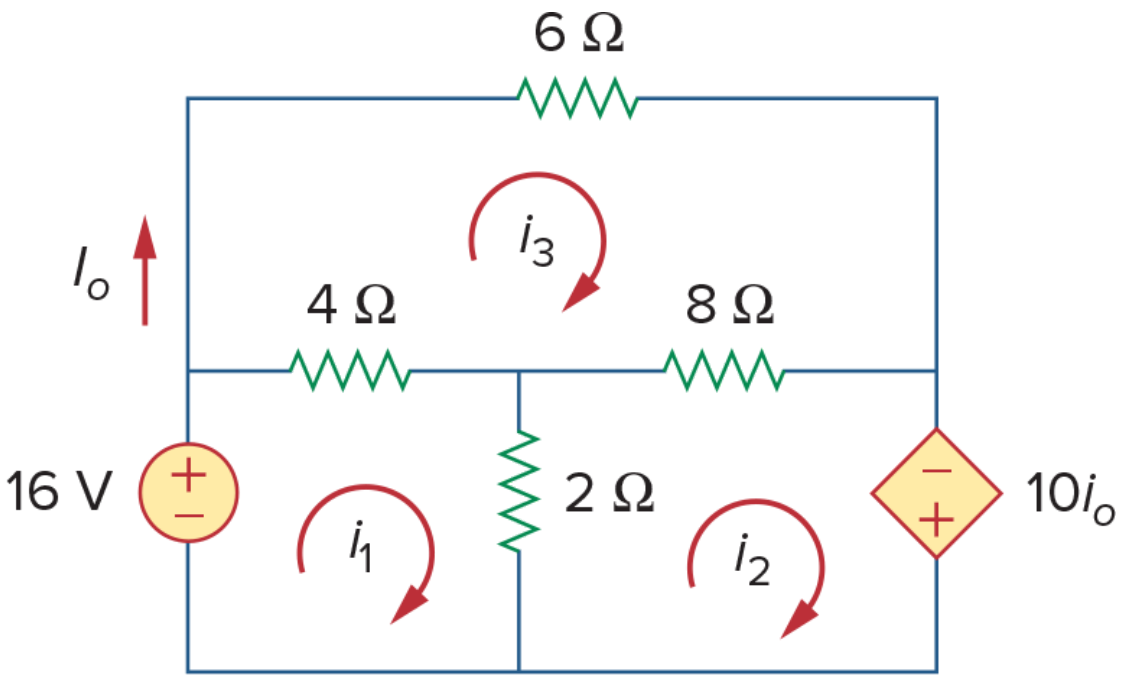



Figure 3.21 For Practice Prob. 3.6.

Answer: – 4 A.

# 3.5 Mesh Analysis with Current Sources

Applying mesh analysis to circuits containing current sources (dependent or independent) may appear complicated. But it is actually much easier than what we encountered in the previous section, because the presence of the current sources reduces the number of equations. Consider the following two possible cases.

▪ CASE 1 When a current source exists only in one mesh: Consider the circuit in Fig. 3.22, for example. We set $i_2 = -5$ A and write a mesh equation for the other mesh in the usual way; that is,

$$-10 + 4i_1 + 6(i_1 - i_2) = 0 \quad \Rightarrow \quad i_1 = -2 \text{ A} \quad \textbf{(3.17)}$$

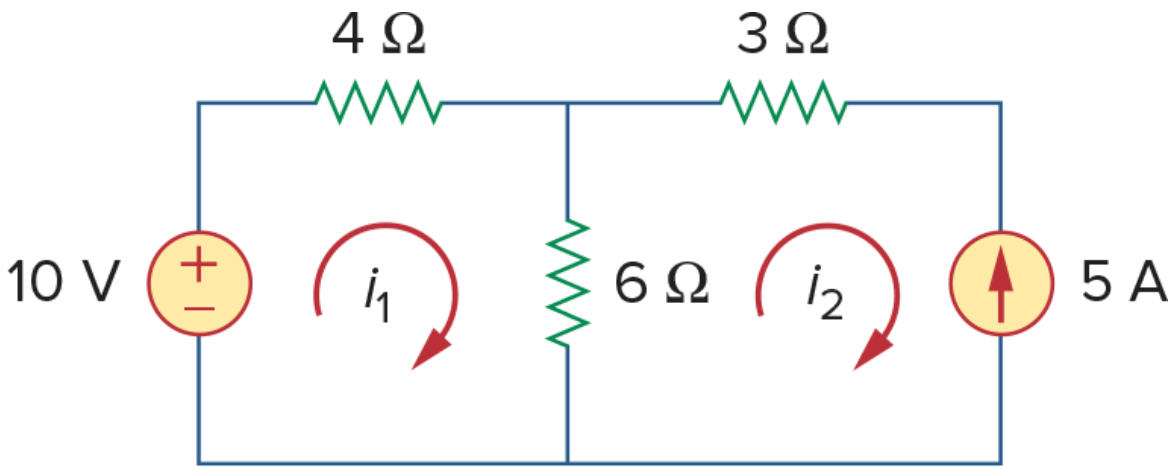
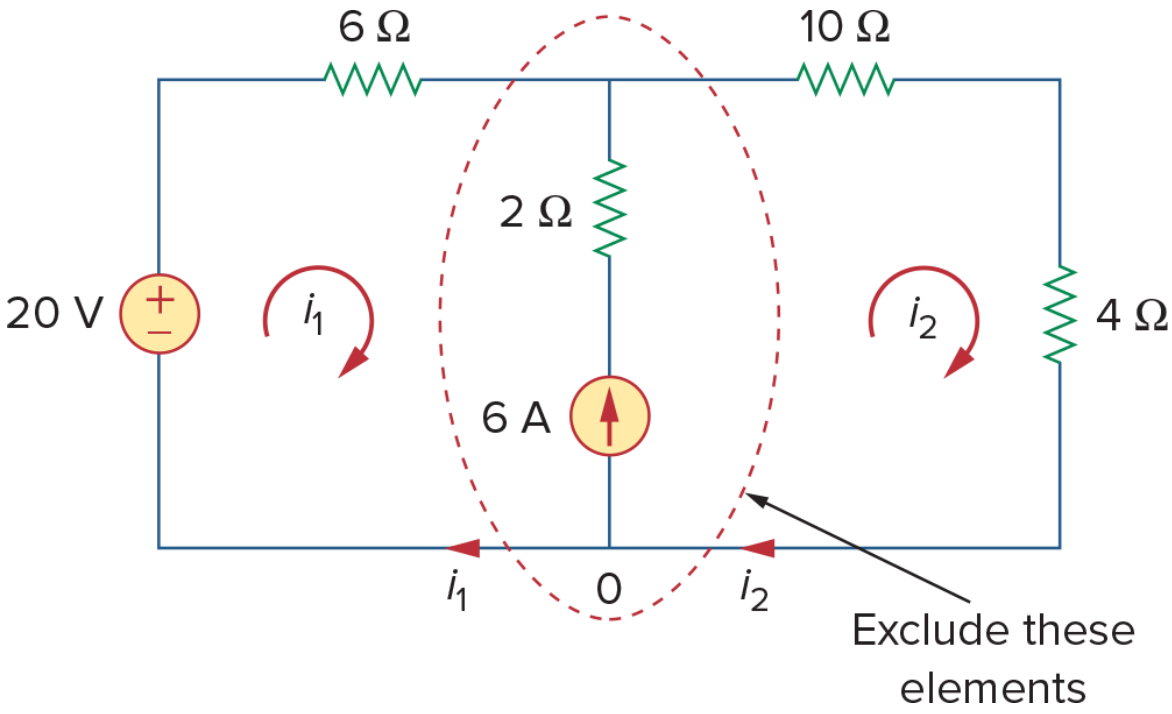


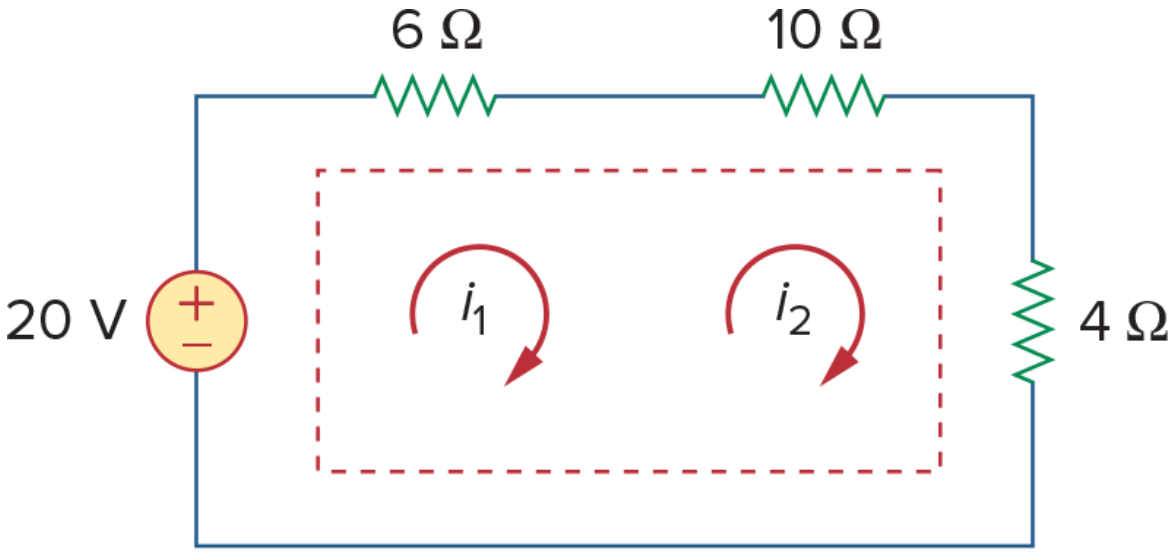

Figure 3.22 A circuit with a current source.

▪ CASE 2 When a current source exists between two meshes: Consider the circuit in Fig. 3.23(a), for example. We create a *supermesh* by excluding the current source and any elements connected in series with it, as shown in Fig. 3.23(b). Thus,

A **supermesh** results when two meshes have a (dependent or independent) current source in common.




(a)




(b)

Figure 3.23 (a) Two meshes having a current source in common, (b) a supermesh, created by excluding the current source.

As shown in Fig. 3.23(b), we create a supermesh as the periphery of the two meshes and treat it differently. (If a circuit has two or more supermeshes that

intersect, they should be combined to form a larger supermesh.) Why treat the supermesh differently? Because mesh analysis applies KVL—which requires that we know the voltage across each branch—and we do not know the voltage across a current source in advance. However, a supermesh must satisfy KVL like any other mesh. Therefore, applying KVL to the supermesh in Fig. 3.23(b) gives

$$-20 + 6i_1 + 10i_2 + 4i_2 = 0$$

Page 97or

$$6i_1 + 14i_2 = 20 \quad \textbf{(3.18)}$$

We apply KCL to a node in the branch where the two meshes intersect. Applying KCL to node 0 in Fig. 3.23(a) gives

$$i_2 = i_1 + 6 \quad \textbf{(3.19)}$$

Solving Eqs. (3.18) and (3.19), we get

$$i_1 = -3.2 \text{ A}, \qquad i_2 = 2.8 \text{ A} \quad \textbf{(3.20)}$$

Note the following properties of a supermesh:

1. The current source in the supermesh provides the constraint equation necessary to solve for the mesh currents.
2. A supermesh has no current of its own.
3. A supermesh requires the application of both KVL and KCL.

**Example 3.7**

For the circuit in Fig. 3.24, find $i_1$ to $i_4$ using mesh analysis.

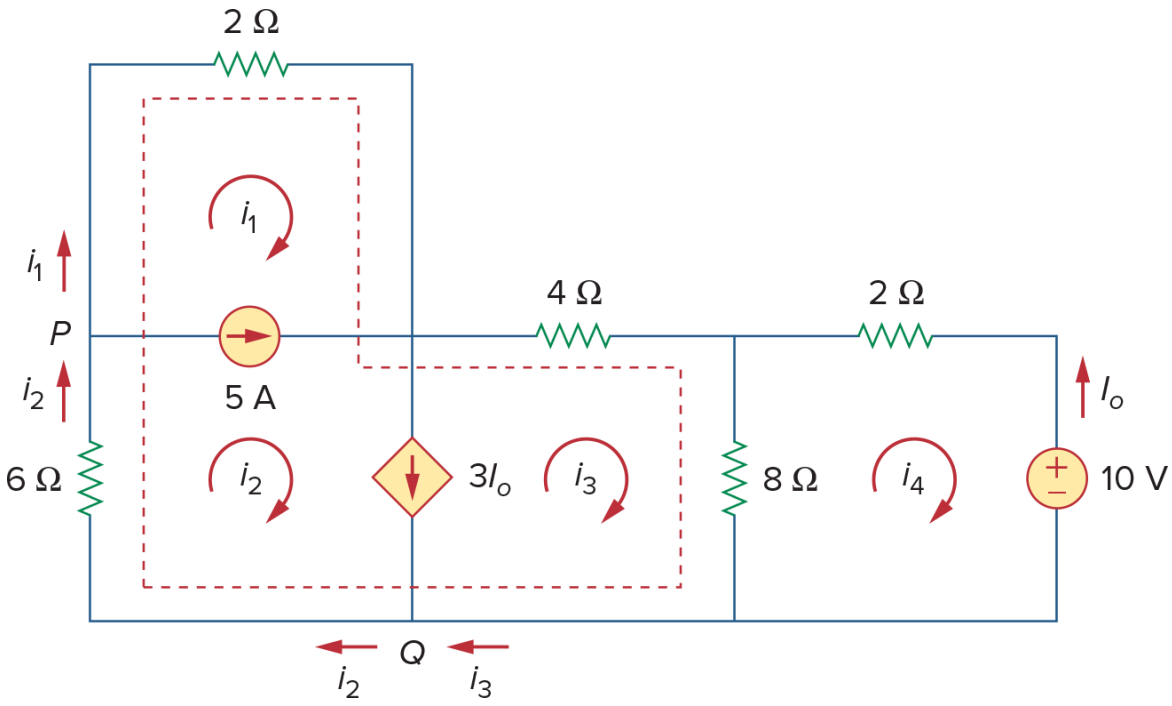



Figure 3.24 For Example 3.7.

**Solution:**

Note that meshes 1 and 2 form a supermesh because they have an independent current source in common. Also, meshes 2 and 3 form another supermesh because they have a dependent current source in common. The two supermeshes intersect and form a larger supermesh as shown. Applying KVL to the larger supermesh,

$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$

or

$$i_1 + 3i_2 + 6i_3 - 4i_4 = 0 \quad \textbf{(3.7.1)}$$

For the independent current source, we apply KCL to node *P*:

$$i_2 = i_1 + 5 \quad \textbf{(3.7.2)}$$

For the dependent current source, we apply KCL to node *Q*:

$$i_2 = i_3 + 3I_o$$

Page 98But $I_o = -i_4$, hence,

$$i_2 = i_3 - 3i_4 \quad \textbf{(3.7.3)}$$

Applying KVL in mesh 4,

$$2i_4 + 8(i_4 - i_3) + 10 = 0$$

or

$$5i_4 - 4i_3 = -5 \quad \textbf{(3.7.4)}$$

From Eqs. (3.7.1) to (3.7.4),

$$i_1 = -7.5 \text{ A}, \qquad i_2 = -2.5 \text{ A}, \qquad i_3 = 3.93 \text{ A}, \qquad i_4 = 2.143 \text{ A}$$

**Practice Problem 3.7**

Use mesh analysis to determine $i_1$, $i_2$, and $i_3$ in Fig. 3.25.

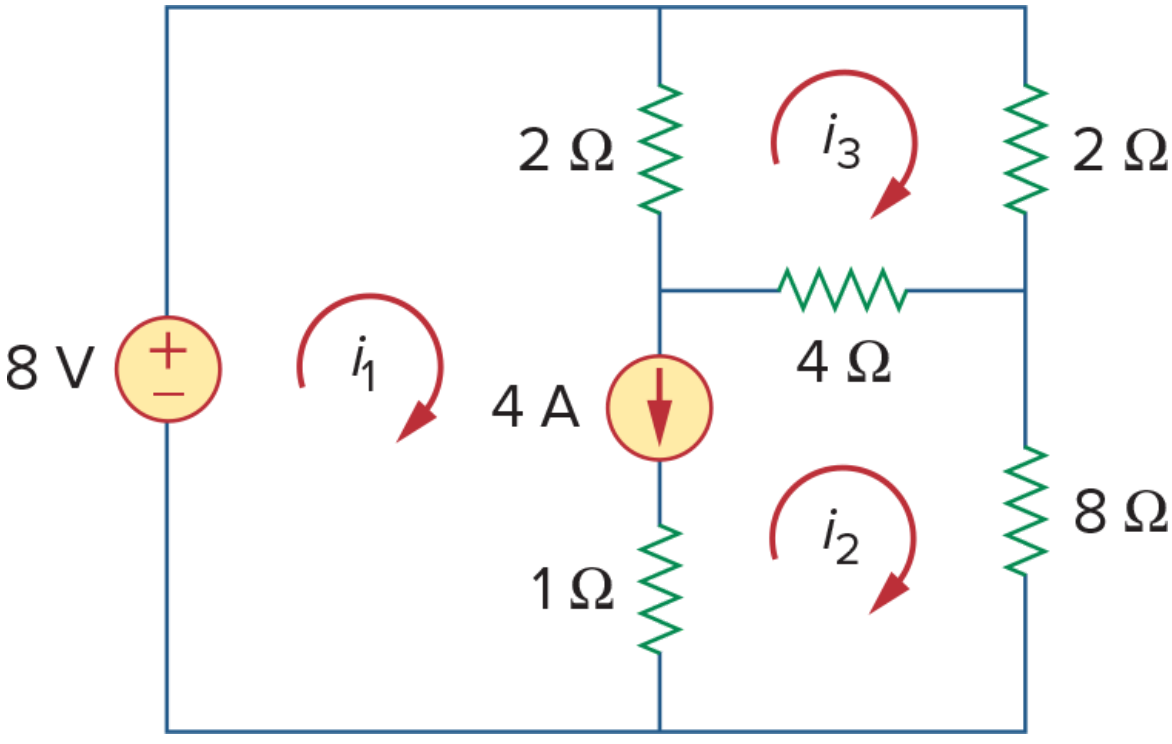



Figure 3.25 For Practice Prob. 3.7.

Answer: $i_1 = 4.632\ \text{A}, i_2 = 631.6\ \text{mA}, i_3 = 1.4736\ \text{A}.$